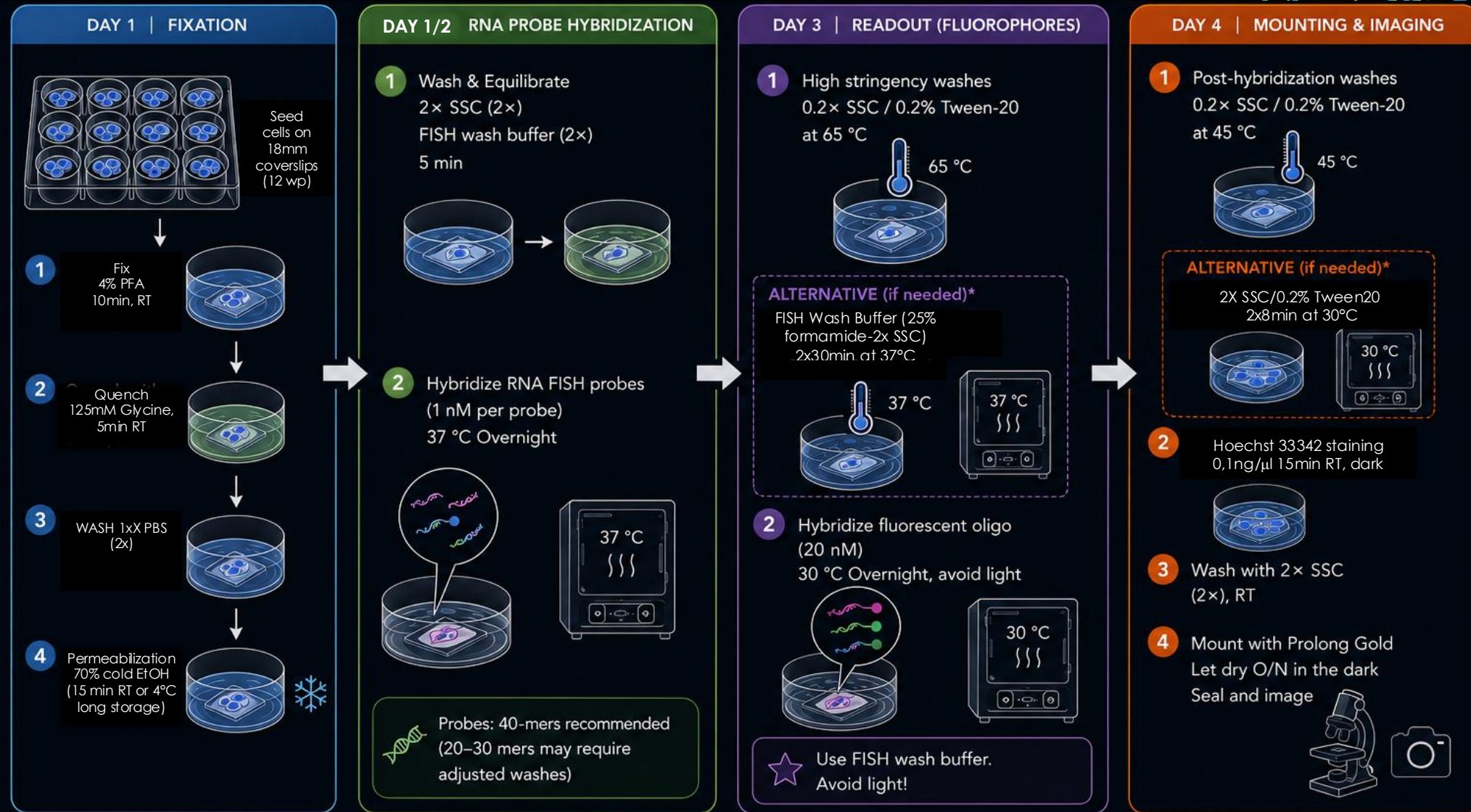


***Fluorescence In Situ Hybridization Methods:
From Theory to Practice***
HT Scientific training course

RNA FISH images analysis by Fiji

Giulia Della Chiara, PhD
Senior Staff Scientist – Bienko Lab

RNA FISH procedure



*Use alternative washes if probes are 20–30 mers long or if signal is low.

How to analyse your images?

nature methods

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Perspective | Published: 28 June 2012

Fiji: an open-source platform for biological-image analysis

[Johannes Schindelin](#), [Ignacio Arganda-Carreras](#), [Erwin Frise](#), [Verena Kaynig](#), [Mark Longair](#), [Tobias Pietzsch](#), [Stephan Preibisch](#), [Curtis Rueden](#), [Stephan Saalfeld](#), [Benjamin Schmid](#), [Jean-Yves Tinevez](#), [Daniel James White](#), [Volker Hartenstein](#), [Kevin Eliceiri](#), [Pavel Tomancak](#) ✉ & [Albert Cardona](#) ✉

[Nature Methods](#) **9**, 676–682 (2012) | [Cite this article](#)

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ImageJ Docs

- Download
- Learn
- Extend
- Contribute
- Discuss
- Explore

Fiji Downloads

Fiji is a distribution of ImageJ which includes many useful plugins [contributed by the community](#).

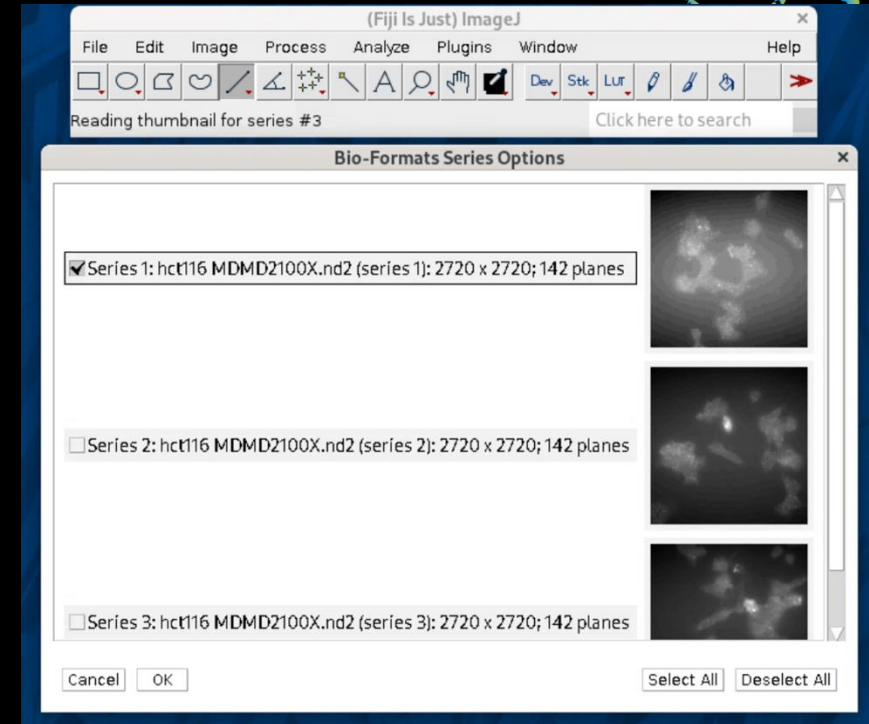
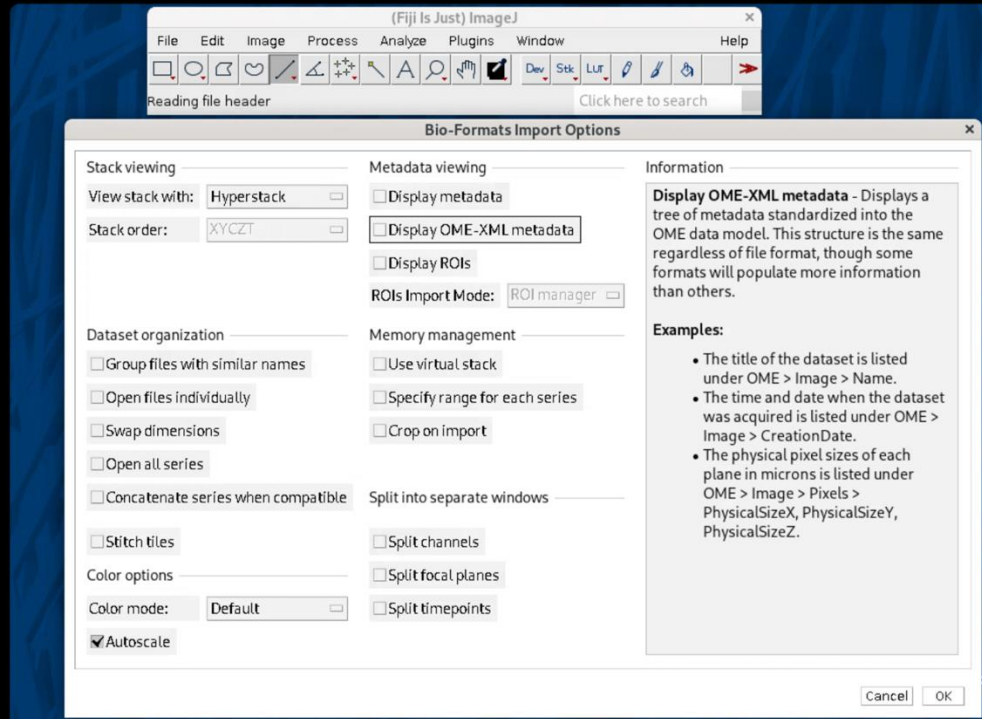
LATEST DOWNLOADS				STABLE DOWNLOADS					
OS	ARCH	DOWNLOAD		OS	ARCH	DOWNLOAD			
Linux	x86-64	US	UK	FR	Linux	x86-64	US	UK	FR
	arm64	US	UK	FR		macOS	x86-64	US	UK
macOS	x86-64	US	UK	FR	Windows	x86-64	US	UK	FR
	arm64	US	UK	FR		x86-32	US	UK	FR
Windows	x86-64	US	UK	FR	Portable (no JDK)		US	UK	FR
	arm64	US	UK	FR					

- Windows XP or later, x86-64 or x86-32
- Mac OS X 10.8 "Mountain Lion" or later in Intel mode
- Linux on x86-64
- Any other system with a Java 8 runtime, except plugins using native libraries (e.g. [3D Viewer](#))

<https://imagej.net/software/fiji/downloads>

<https://imagej.net/ij/download.html> (Java build)

Upload and open your image on Fiji

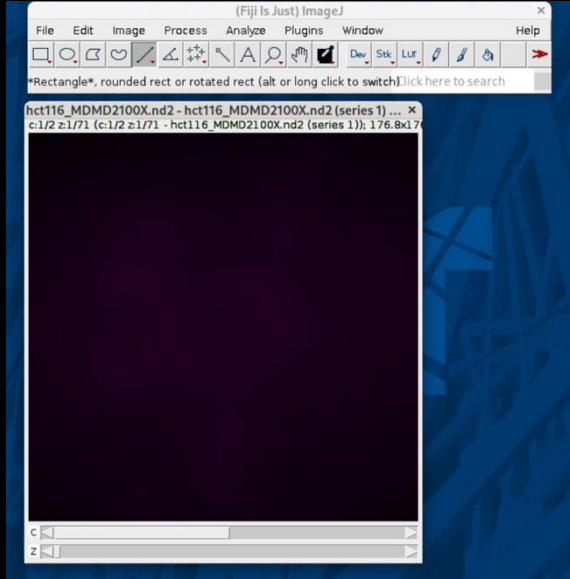


Hyperstack modality preserves *the biological organization* of the acquisition:

- channels (C)
- z slices (Z)
- time points (T)*

*If present

Image physical properties



- Pixel size

- Voxel depth (step size)

<https://svi.nl/Nyquist-Calculator>

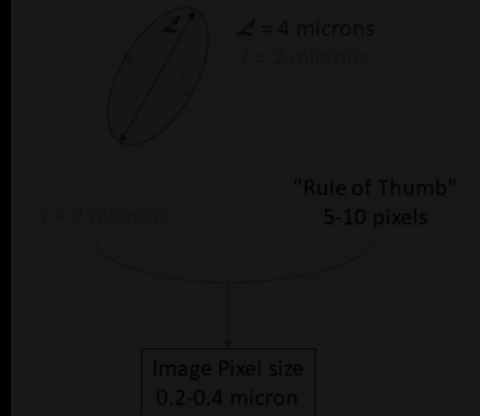
- z spacing (#steps)

- Bit depth (16- bit)

Rule of Thumb



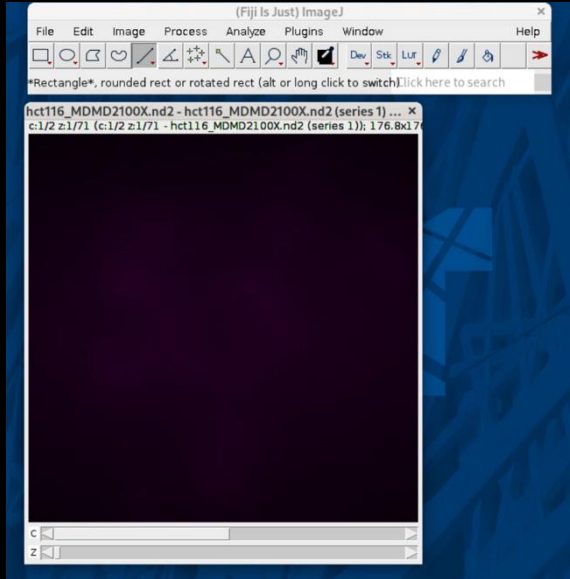
Example



If your object size is below the limit of diffraction (see [Abbe diffraction limit](#), i.e. roughly 250 nm), there is no need to acquire images with pixel size of 2 Å !

Adapt **Pixel Size** of your image to the smallest object of interest in your sample (and to the microscope capability)

Image physical properties



- Pixel size

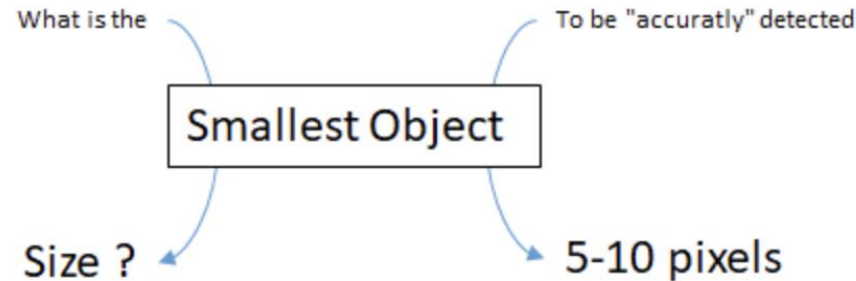
- Voxel depth (step size)

<https://svi.nl/Nyquist-Calculator>

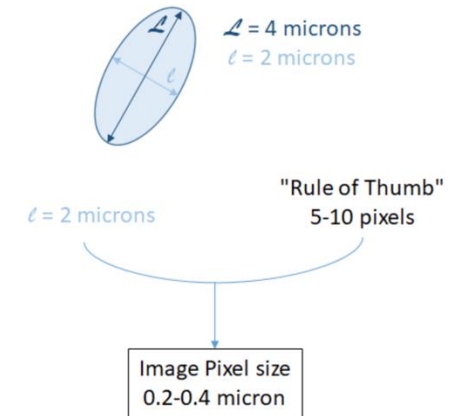
- z spacing (#steps)

- Bit depth (16- bit)

Rule of Thumb



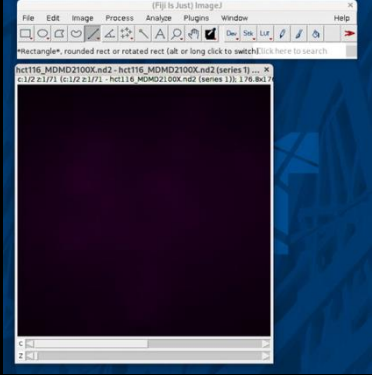
Example



If your object size is below the limit of diffraction (see [Abbe diffraction limit](#), i.e. roughly 250 nm), there is no need to acquire images with pixel size of 2 Å !

Adapt **Pixel Size** of your image to the smallest object of interest in your sample (and to the microscope capability)

Image physical properties



- Pixel size

- Voxel depth (step size)

<https://svi.nl/Nyquist-Calculator>

- z spacing (#steps)

- Bit depth (16- bit)

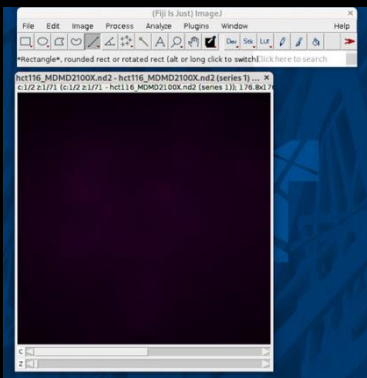
3D equivalent of a pixel

Voxel = size in X/Y/Z (3D)



Pixel width x Pixel height x Z spacing

Image physical properties



bit depth	stored values	range
8	256	0 to 255
12	4096	0 to 4095
16	65536	0 to 65535

- Pixel size

- Voxel depth (step size)

<https://svi.nl/Nyquist-Calculator>

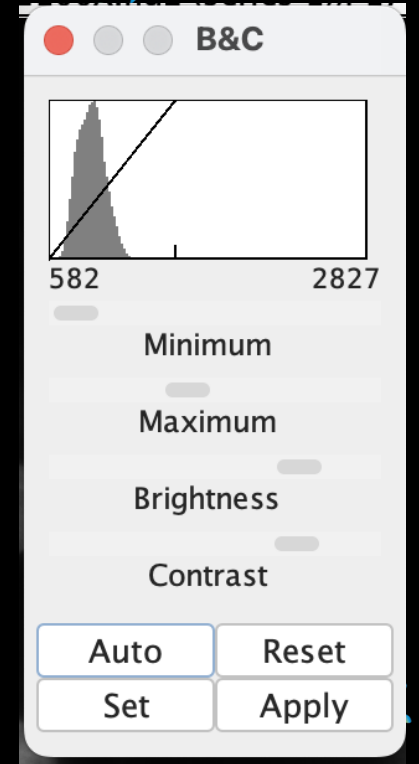
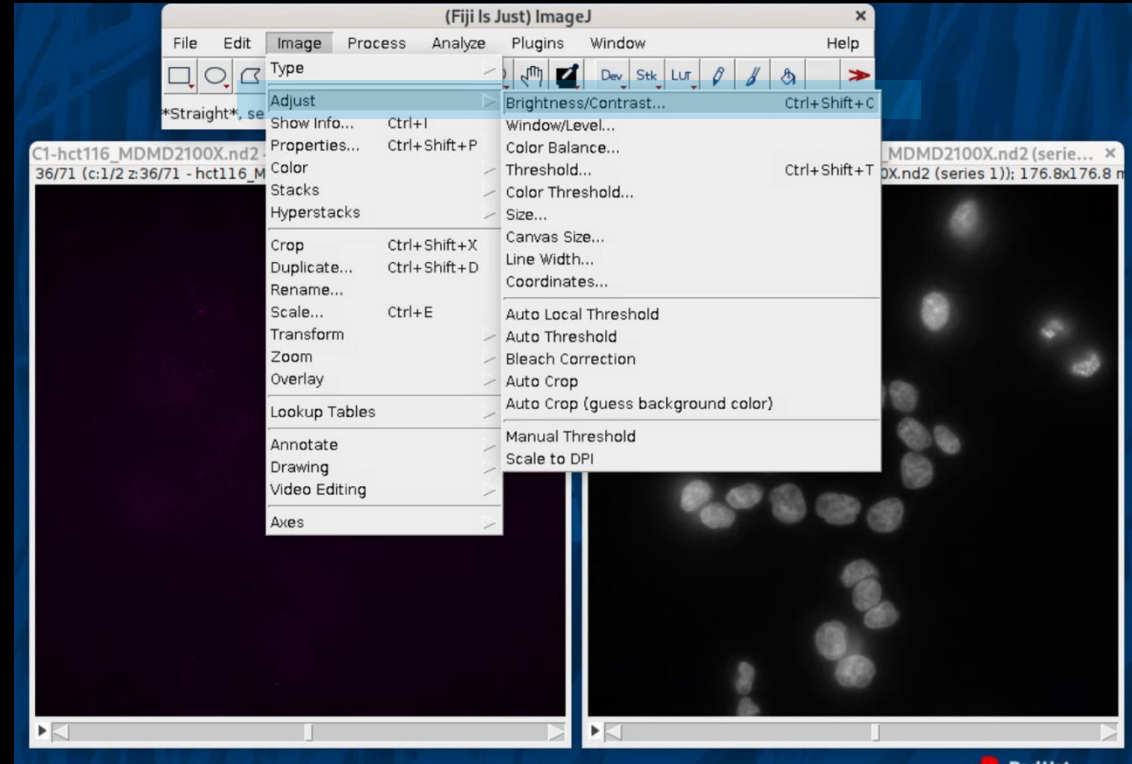
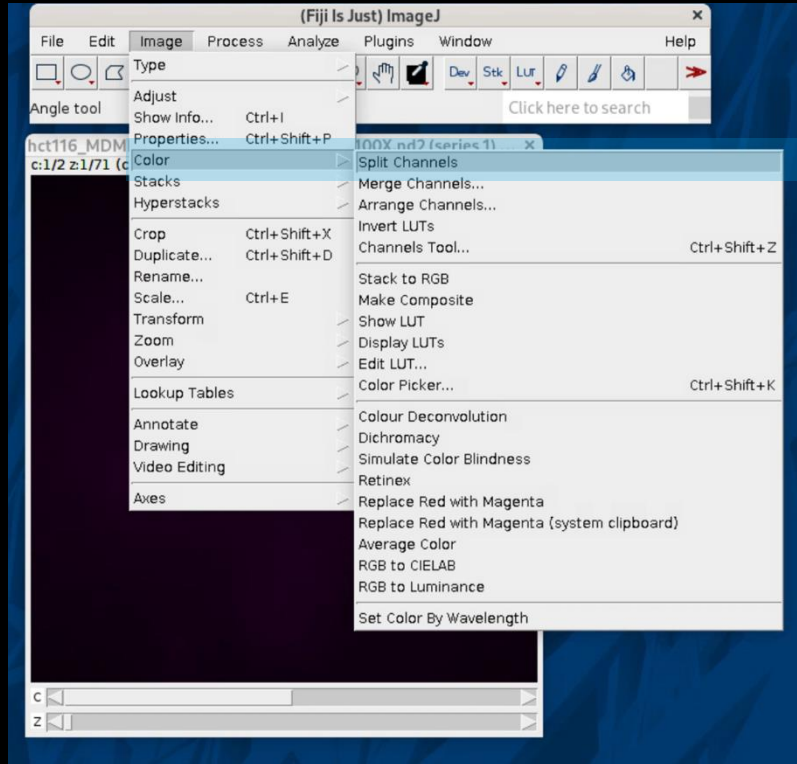
- z spacing (#steps)

- Bit depth (16- bit)

DO NOT SATURATE your images !

The more values you have at your disposal, **the more you can discriminate the high and low intensities without saturating pixels nor losing them** within the background.

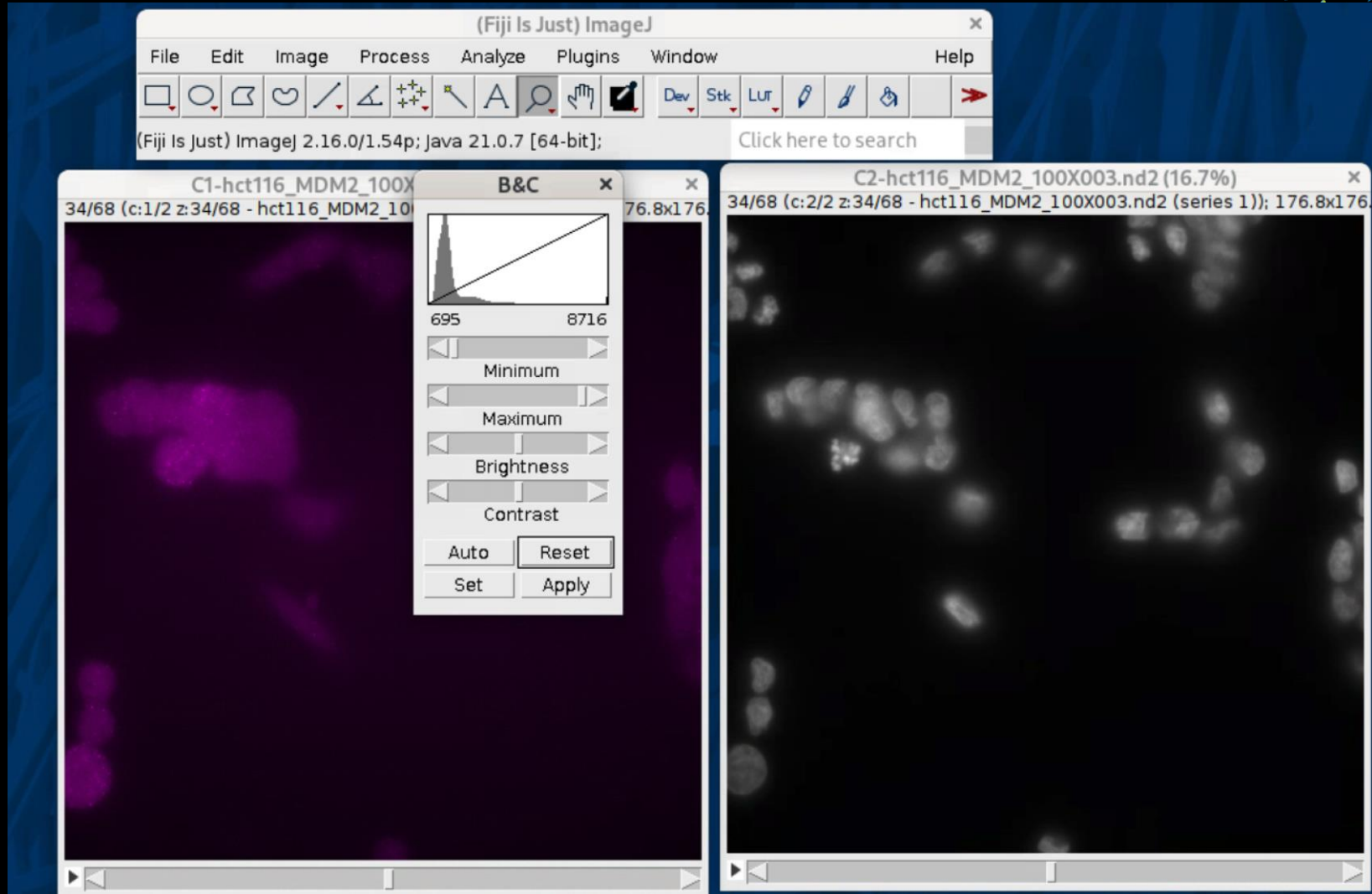
Split the colors and adjust the brightness and contrast (I)



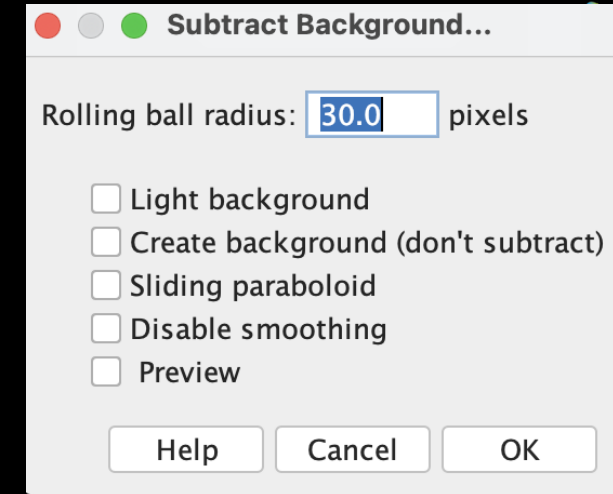
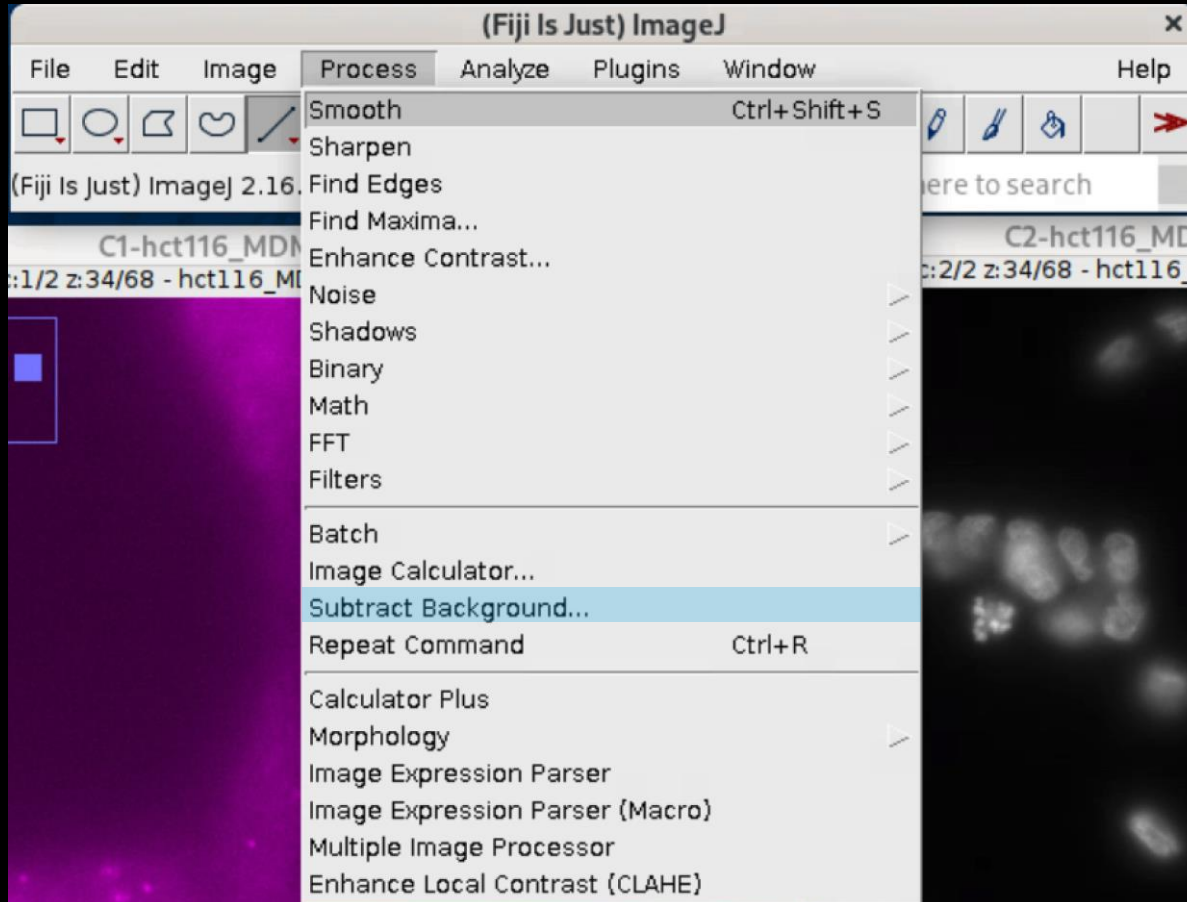
Each fluorophore channel contains:

- different signal distributions
- different background
- different noise levels

Split the colors and adjust the brightness and contrast (I)



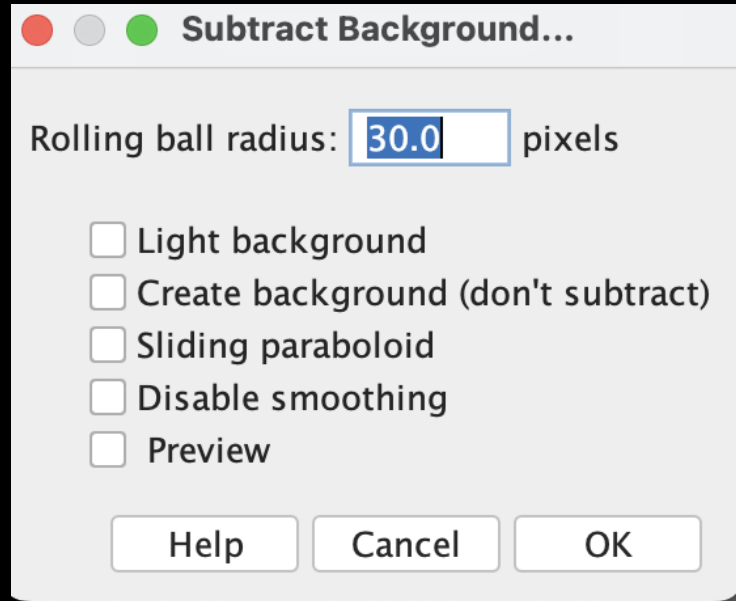
Application of background correction: Subtract background



radius should be at least **as large as the largest object** that is NOT background.

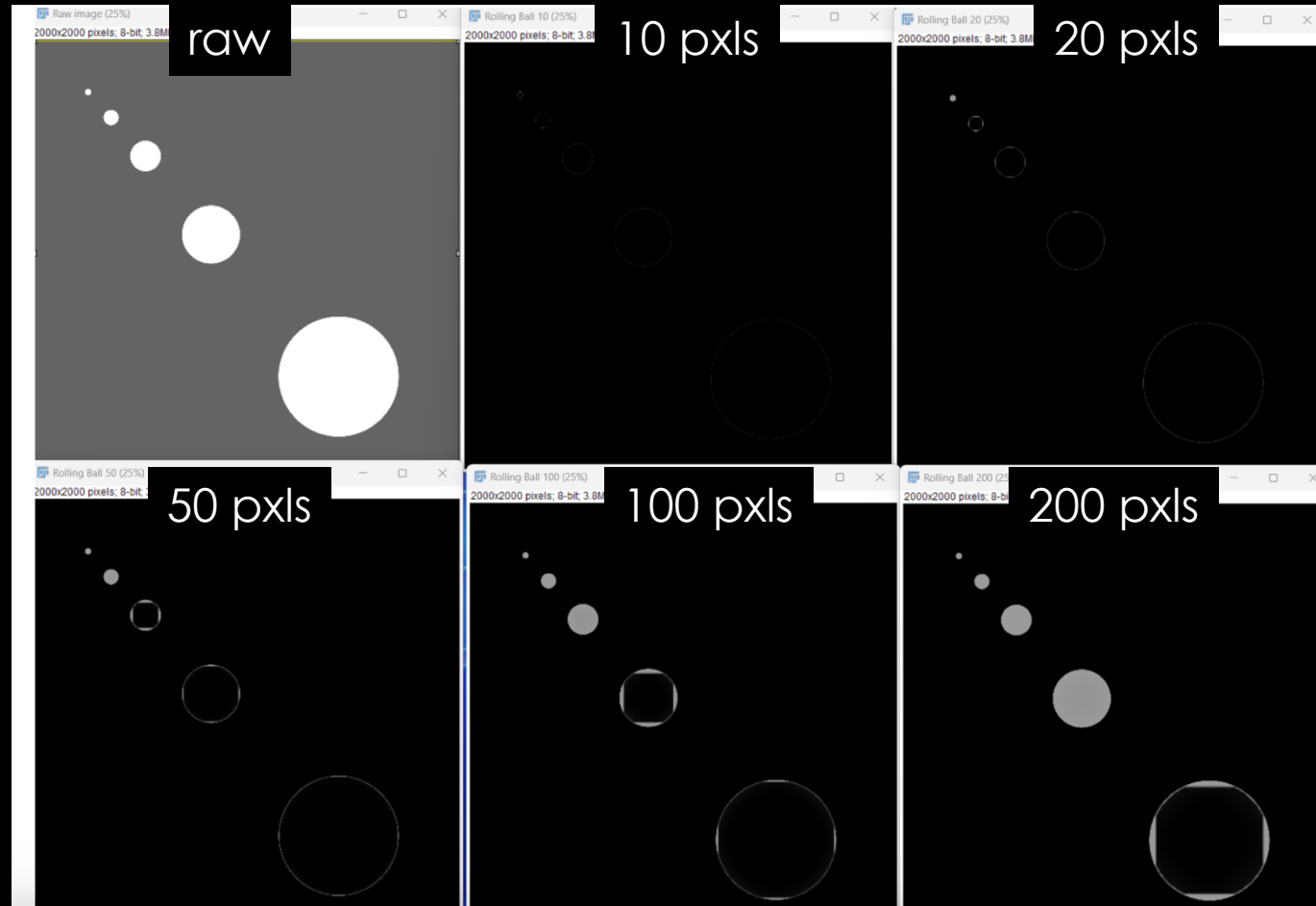
SOURCE: [https://imagejdocu.list.lu/gui/process/subtract_background?s\[\]=background](https://imagejdocu.list.lu/gui/process/subtract_background?s[]=background)

Subtract background different features

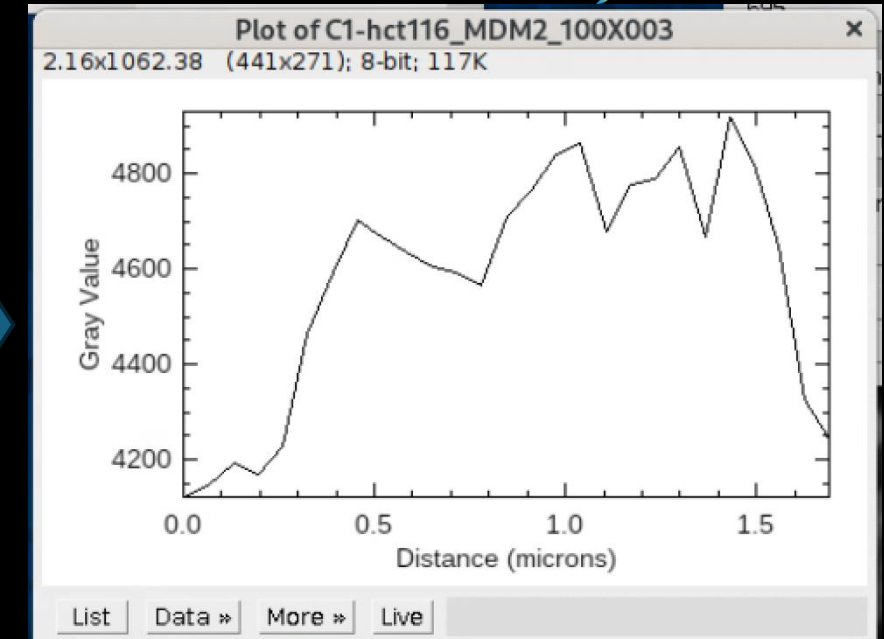
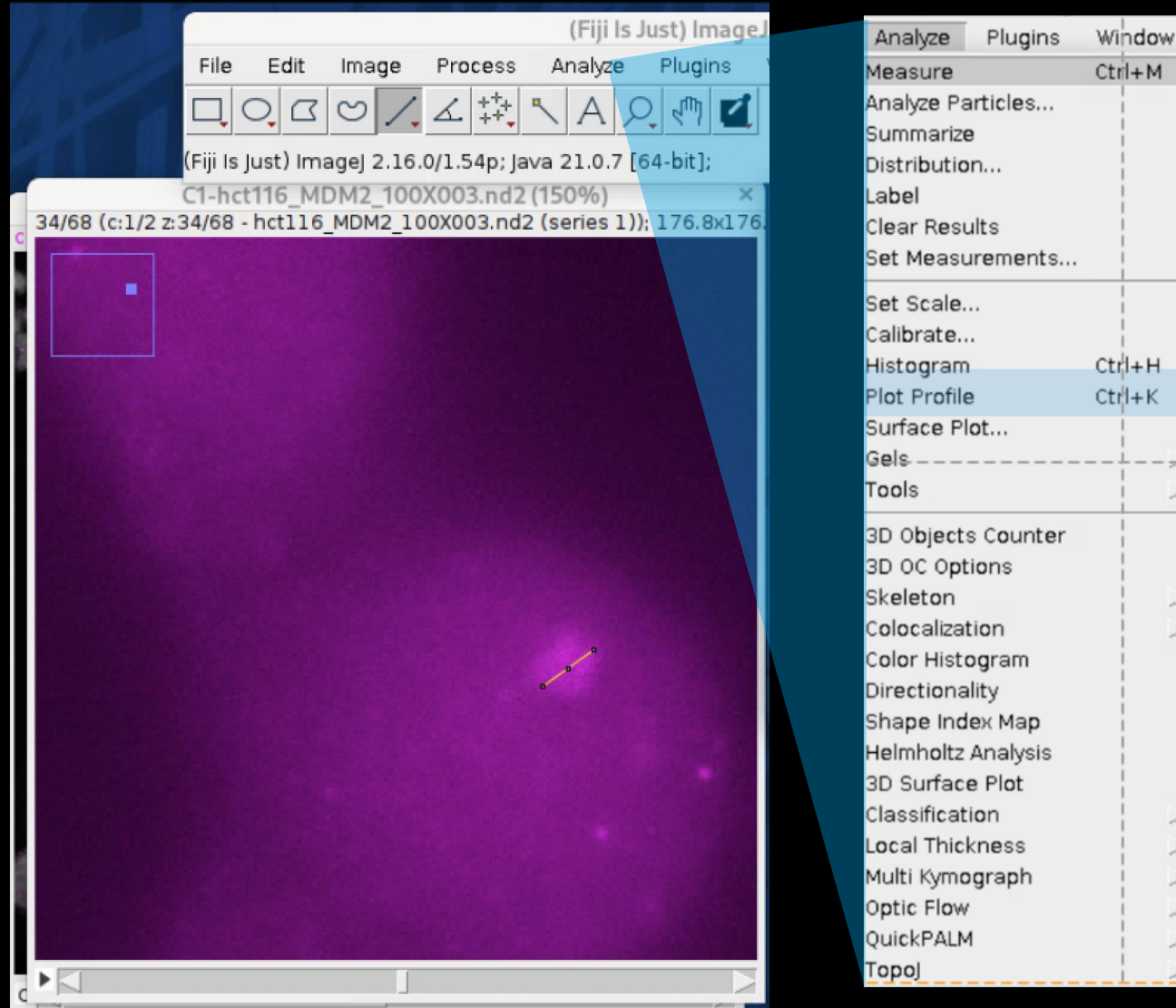


- Light background: dark background bright objects
- Create background: output is the background
- Sliding paraboloid: it replaces the rolling ball with a paraboloid of the same curvature
- Disable smoothing: 3×3 averaging filter before background estimation

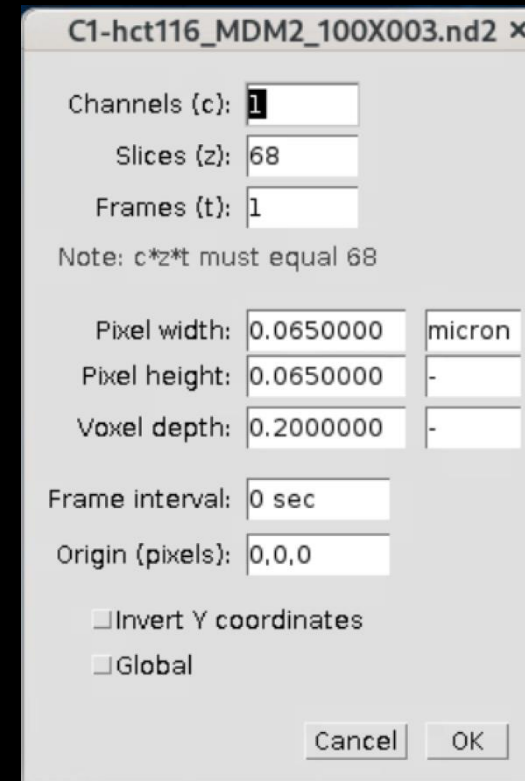
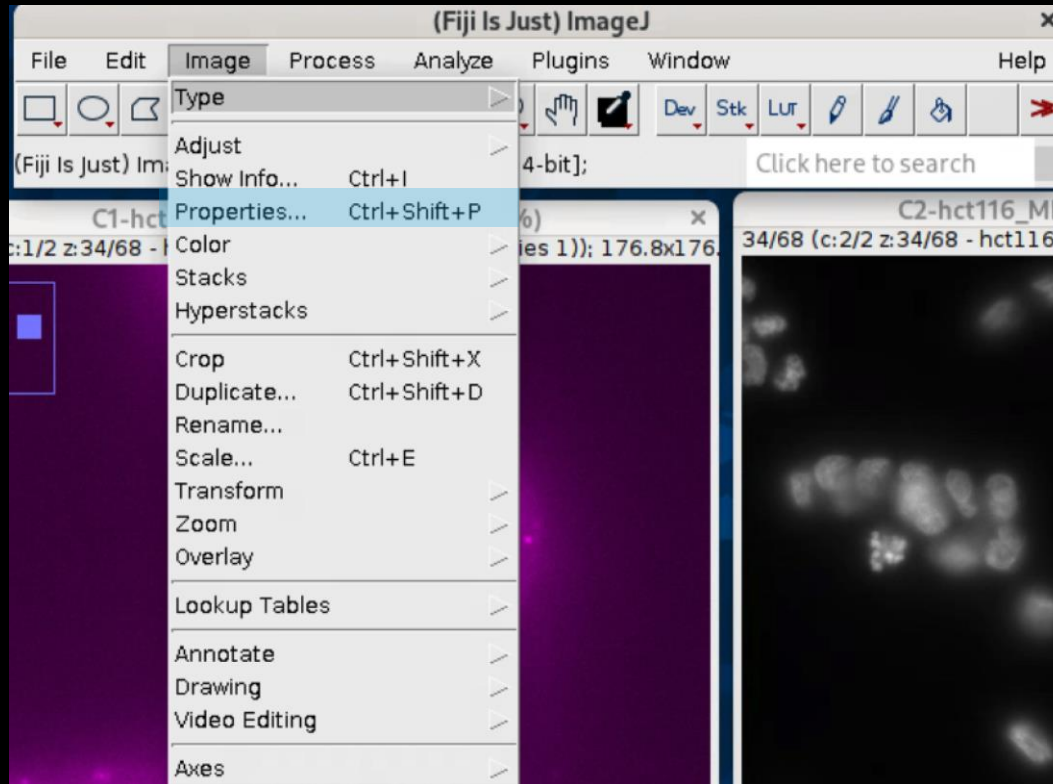
Application of correct rolling bar radius to remove background and keep all your objects



Largest object calculation (I)



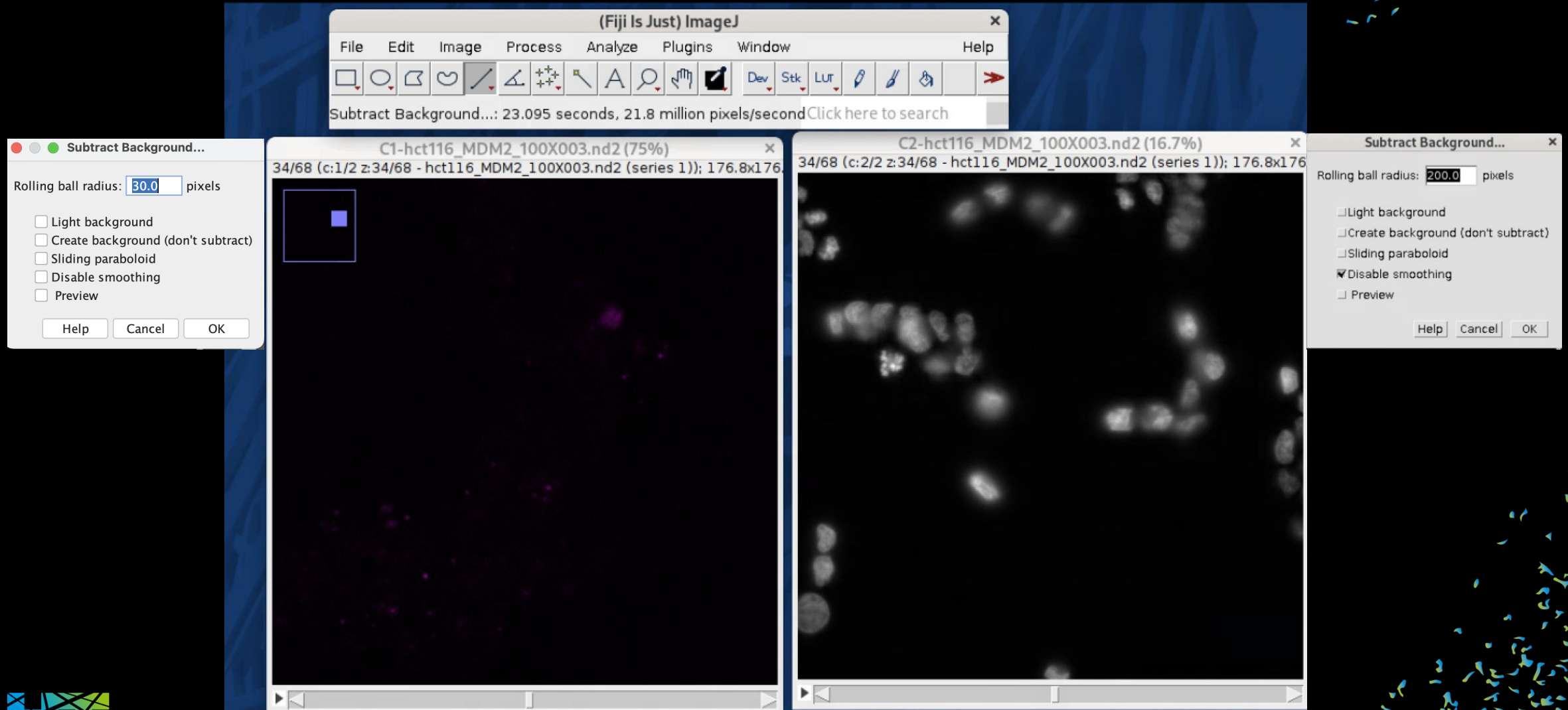
Largest object calculation (II)



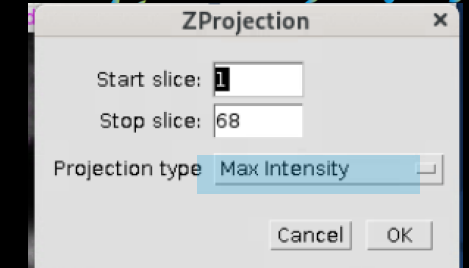
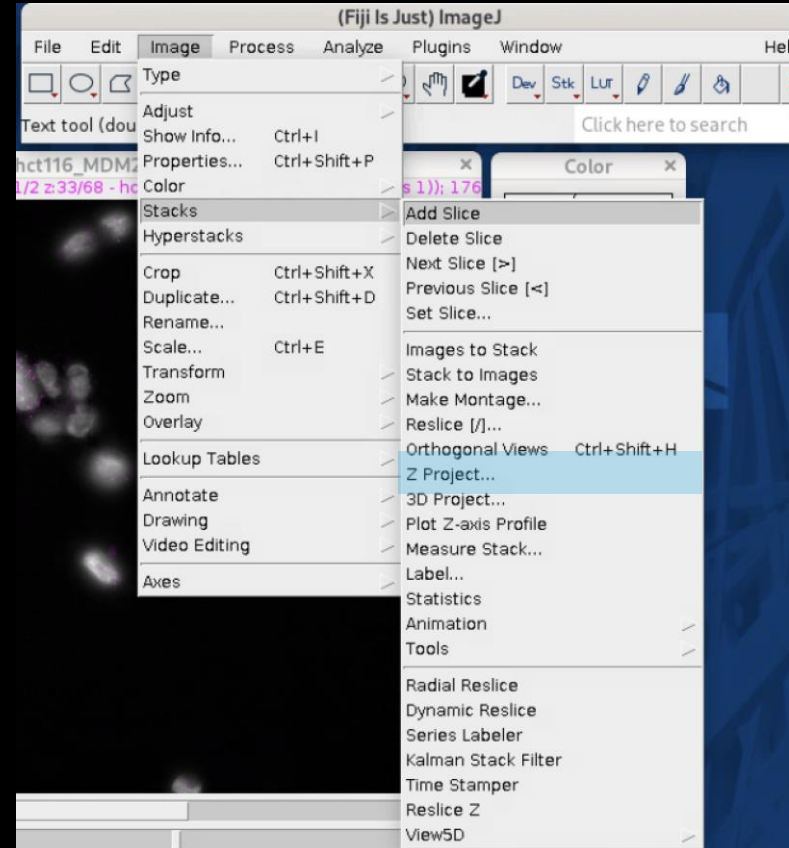
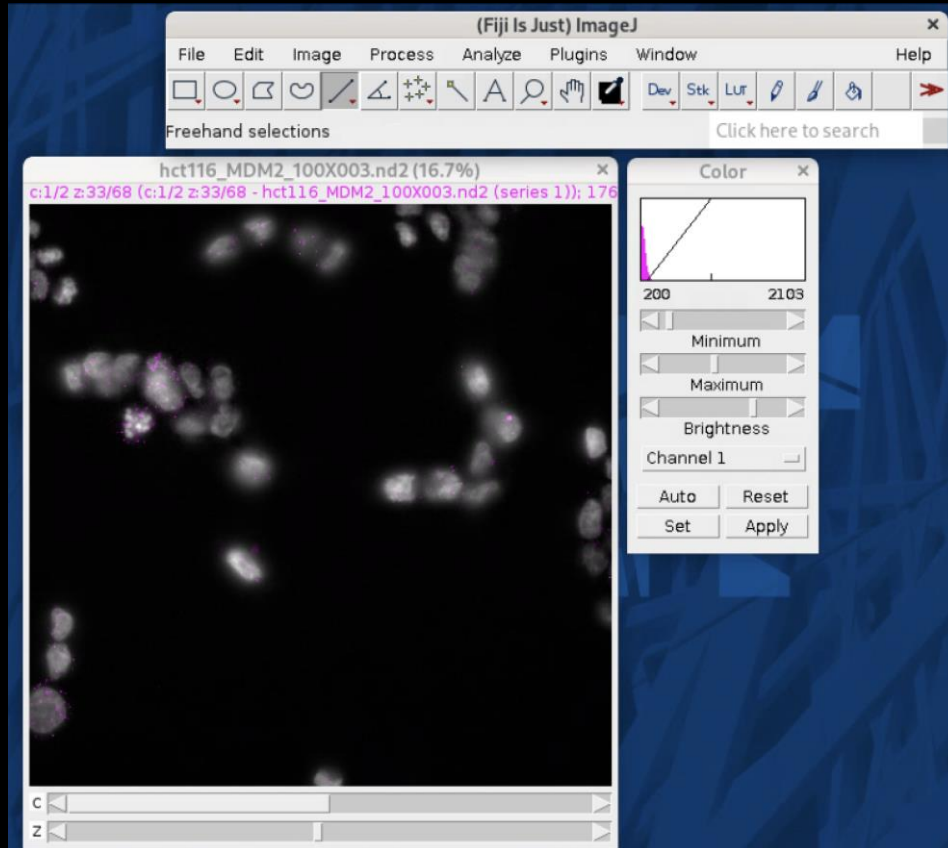
1 pixel = 0,065 μ m

$$1 \text{ pxl: } 0,065 \mu\text{m} = x_{\text{pxls}}: 1,67 \mu\text{m}$$
$$X_{\text{pxls}} = 1,67 \mu\text{m} / 0,065 \mu\text{m} = 25,7$$

Application of background correction choosing the right rolling bar radius

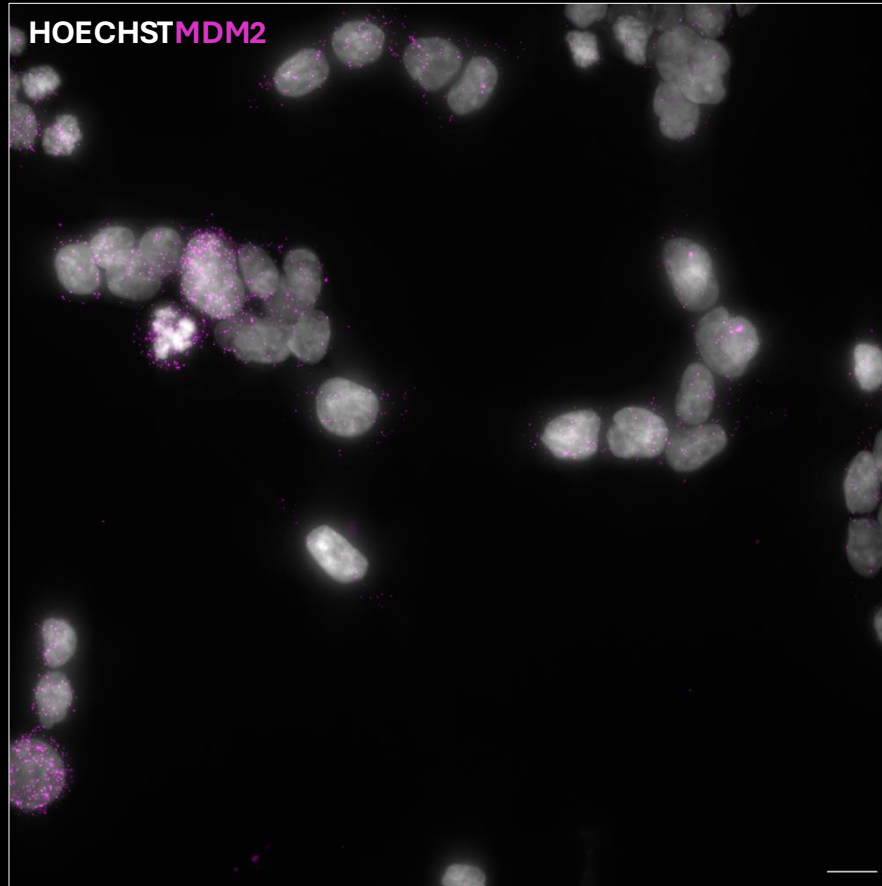


Merge channels and stacks

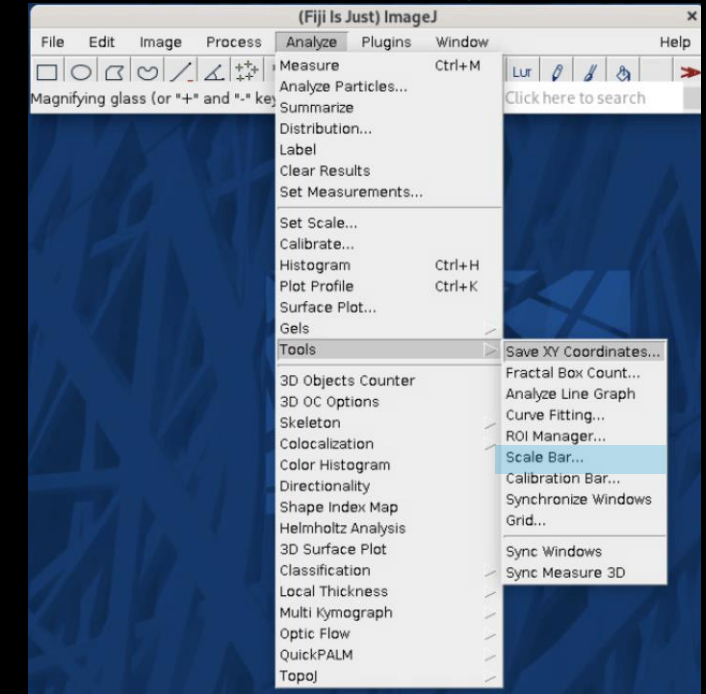
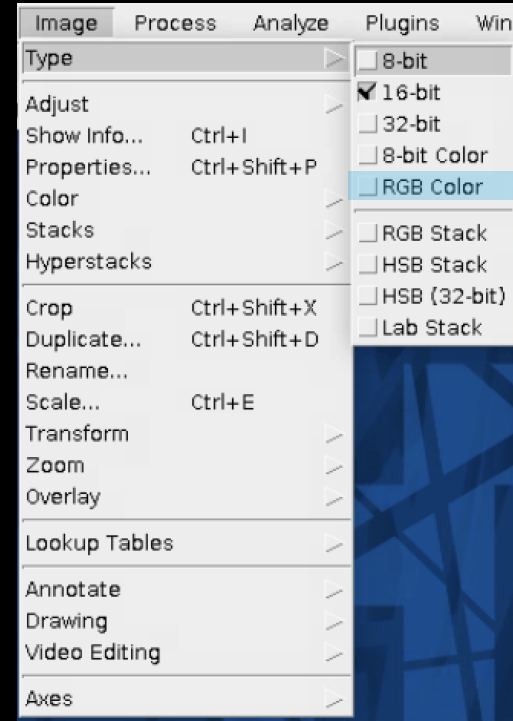


Max projections compress 3D → 2D
The output contains all the pixels with highest intensity for each stack

Save the max intensity projection image and apply scale bar



Scale bar 10um
MaxP: 68 steps every 0,2um step size

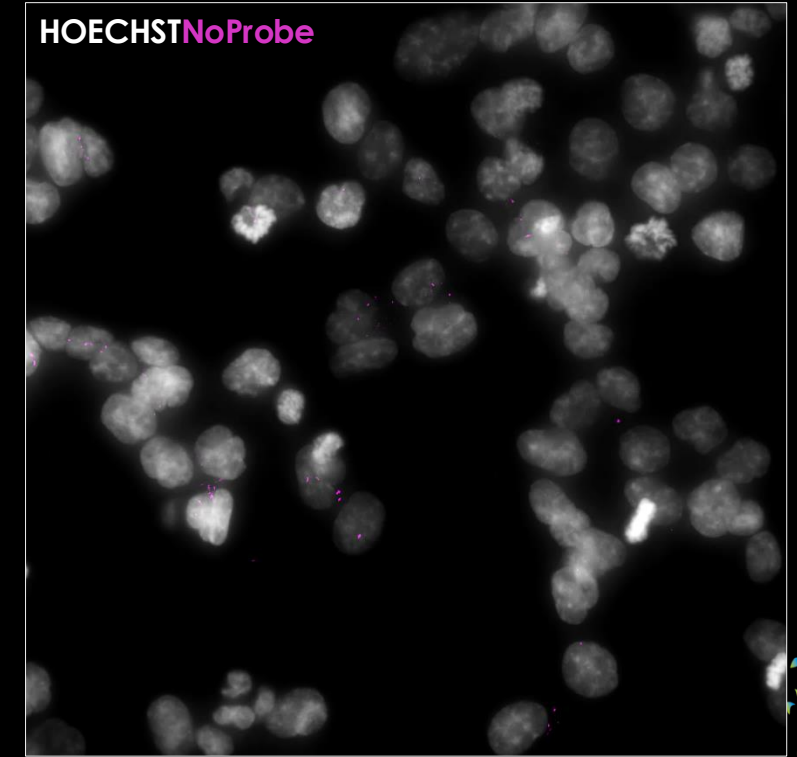
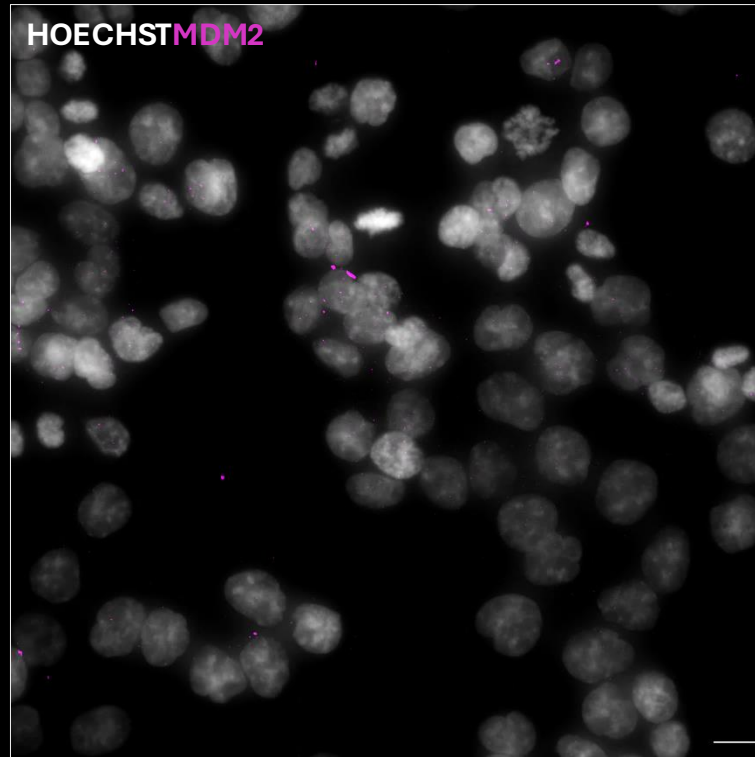


To export images in ppt convert in RGB and then apply scale bar

Save first a TIFF format to preserve all metadata

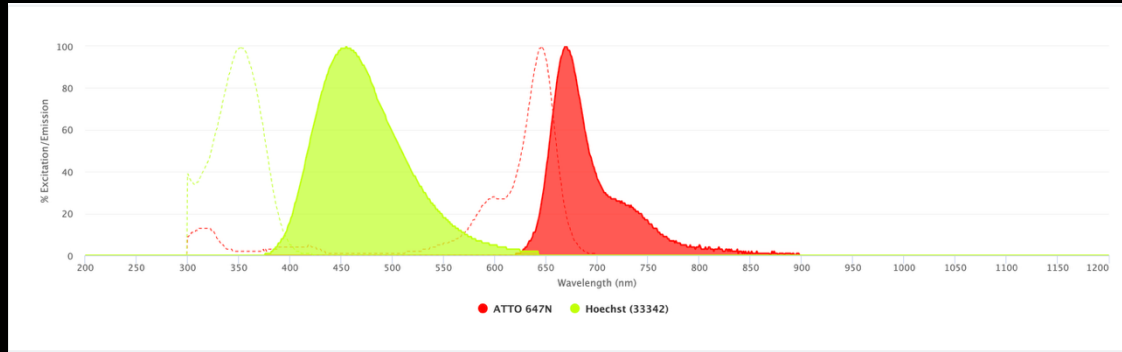
Representative results of HT FISH course practical exercise

+ Rnase A (15min - 37°C)



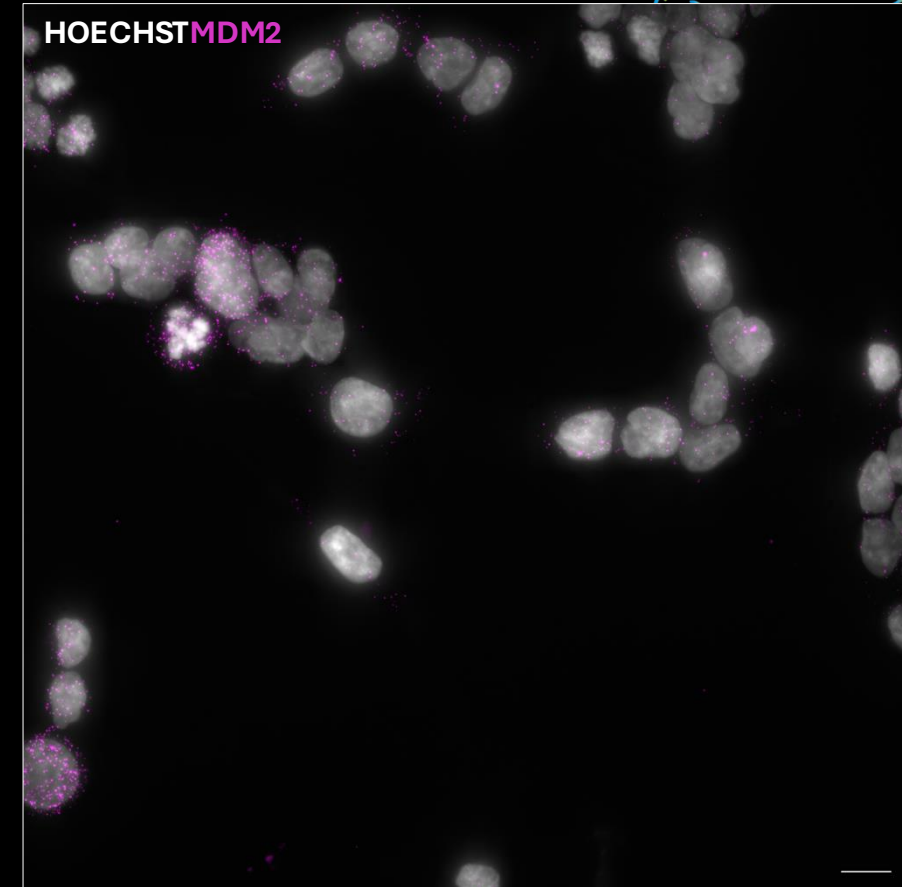
Scale bar 10um
MaxP: 71 steps every 0,2um step size
ATTO647N - intensity settings: 486 - 1100 range
HOECHST 375 settings: 2682 - 25000 range

Maximum projection of final image



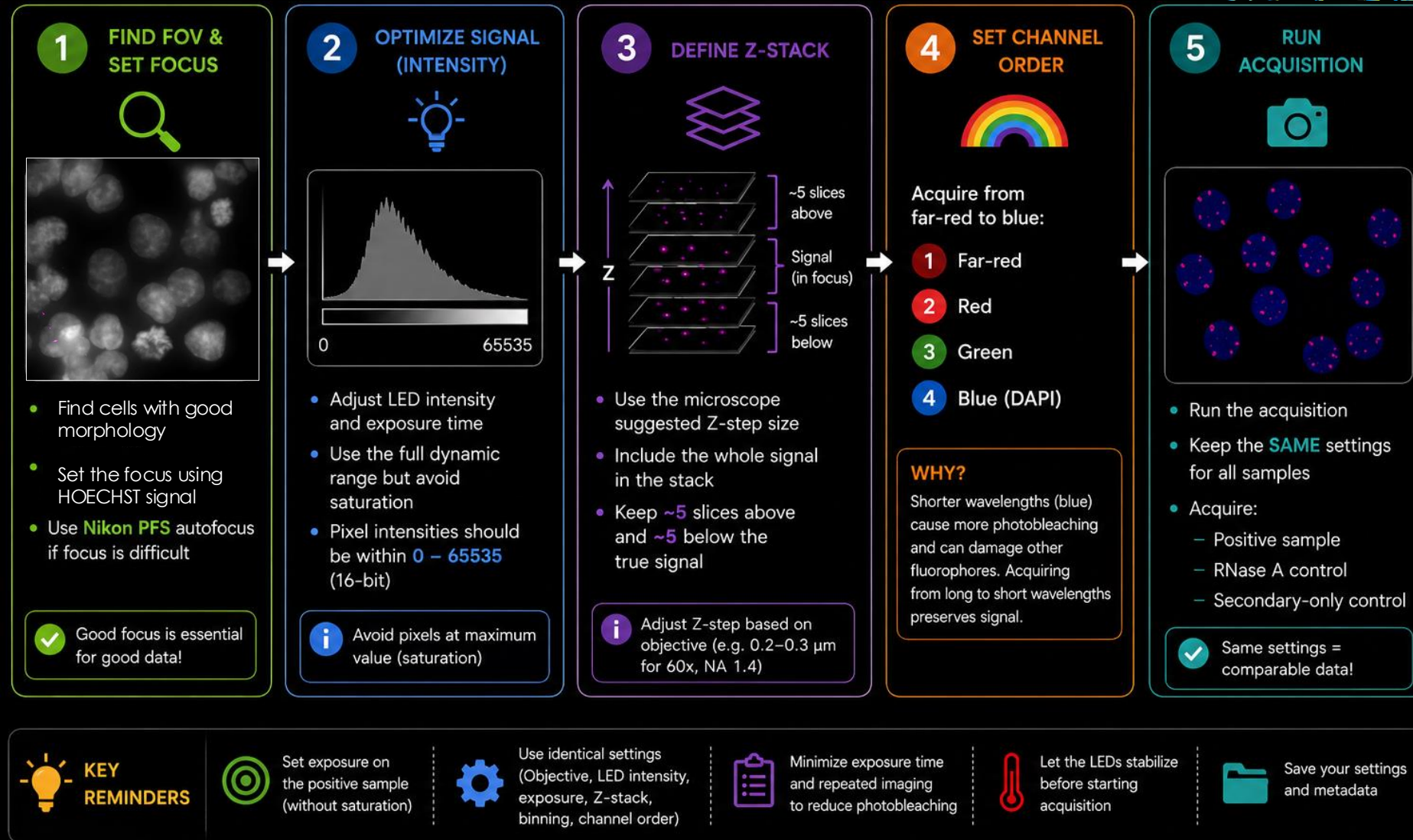
Channel	Filter cube/ band	Led Intensity (%)	Emission (λ)	Time of exposure (ms)	Assigned LUT
1	QB375	7,9	455	25	gray
2	QB630	20	675	800	magenta

Nikon Ti2 – widefield modality
Objective: 100x oil, NA: 1,45
Acquisition in 16-bit, ROI 2048x2048, z-step 0,2um for 14um



Scale bar 10um
MaxP: 68 steps every 0,2um step size

RNA FISH imaging acquisition workflow



To correct for chromatic aberrations and shifts between channels, use TetraSpeck Microspheres (0.1 μm , fluorescent blue/green/orange and dark red -Thermo Fisher Scientific, cat. no. T7279)
Before or after each imaging session

FISH images acquisition (II)

✓ WHY WIDEFIELD WITH LED?

- Fast acquisition
- High sensitivity for dim RNA spots
- Gentle illumination → less photobleaching
- Large field of view
- Ideal for thin, fixed samples and transcript counting



WIDEFIELD vs CONFOCAL (SPINNING DISK)

	WIDEFIELD	CONFOCAL
 Speed	Faster	Slower
 Sensitivity	More sensitive (weak RNA spots)	Less sensitive
 Background	More background	Cleaner images (better background rejection)
 Best for	Thin samples, 2D, counting	Thick samples and 3D
 Photobleaching	Lower risk	Higher risk



For most RNA FISH on fixed cells → widefield is the best choice!

LED vs LASER

LED



- Broad spectrum
- Lower intensity
- Gentle on samples
- Less photobleaching
- Stable and easy to use
- Ideal for widefield RNA FISH

LASER



- Single wavelength
- High intensity
- Strong excitation
- Better for confocal/high-speed imaging
- More photobleaching risk

STARTING FISH ACQUISITION

: 🐟 🐟 — Finding those transcripts! — 🐟 🐟 :



KEY REMINDERS



Set exposure on positive samples (no saturation)



Use same settings across all samples (objective, led %, exposure, Z-stack, channel order)



Minimize exposure time and repeated imaging to reduce photobleaching



Switch on LEDs before the acquisition to stabilize them



Save your settings and metadata



Sources: <https://imagejdocu.list.lu/start>
<https://wiki-biop.epfl.ch/humantechnopole.it>